

My Understanding of Survival Strategies in the Malaria Parasite *Plasmodium falciparum*

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A parasite-encoded metabolic pathway allows *Plasmodium falciparum* to take possession of human erythrocyte resources. The mimicry in the environment also prevents immunological pressure and it also causes mutations which leads to pharmaco-resistance. Investigating the heme biosynthetic pathway of this parasite has suggested that it is also capable to sequester significant quantities of hemes from host haemoglobin degradation into a parasite digestive vacuole and it still have genes coding for enzymes necessary to perform de novo synthesis of heme..

The saddling of δ -aminolevulinic acid dehydratase shows that genome of the parasite encodes functional enzyme (PfALAD) which translates into an octameric enzyme that exhibits optimal catalytic activity at alkaline pH. This enzyme is in turn transported to the plastid-like organelle, Apicoplast. In contrast, a quantitative assay of enzyme activity indicated that the parasite-derived PfALAD contributed to only a minor fraction of the total ALAD present in infected red blood cells. Experimental data suggest that host ALAD plays a predominant role in total activity within infected cells. The *Plasmodium berghei* model system, showed that host erythrocyte ALAD(+) binds specifically to its membrane proteins. Disruption of this binding interaction by competitive peptide fragments led to reduced ALAD activity, decreased heme synthesis, and inhibition of parasite growth. This does not sustain the parasite in the host. As an example, mutations in the *Pfcr* are associated with strong chloroquine resistance.

Wide selection of chloroquine-resistant plasmodium populates were demonstrated by high prevalence of K76T mutation and SVMNT haplotype. Integral to the persistence of malaria parasites through a coherent mutational strategy within drug receptor binding pockets. It is evident that survival strategies are in metabolism and evolution, and these potential chinks could be the focus of drug design in future. The effect of mutations on ligand-binding, stability and interaction patterns can be summarized by docking chloroquine or similar molecules into the protein structure.

Theoretically, complementary methods such as molecular dynamics simulations and membrane transport pathway observables will help explain how these mutations modify the flux landscape.

References:

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